

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458871.8018 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.031
Transit depth (Rp/Rs)^2	0.91 +/- 0.59 [%]
Orbital Inclination (inc)	85.4 +/- 1.84
Airmass coefficient 1 (a1)	1.076 +/- 0.01
Airmass coefficient 2 (a2)	-0.069 +/- 0.013
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	#2 - [220, 182]
Optimal Aperture	6.76
Optimal Annulus	15.96
Transit Duration (day)	0.052 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.096 ± 0.031 vs 0.1572 (deviation 1.97σ)
Duration fit vs published	0.052 d vs 0.075 d (deviation 1.44σ)
Mid-time O-C	-5.2 min
Depth SNR	3.1
Residual scatter	0.97 %

Light curve

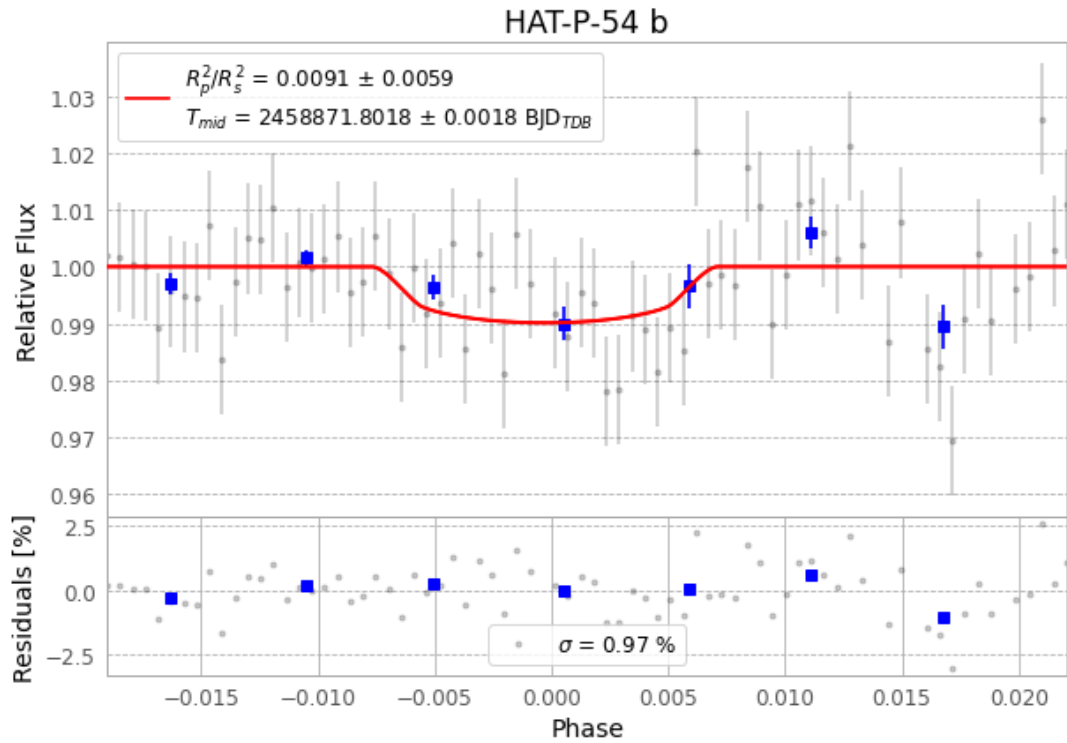


Figure 1 — FinalLightCurve_HAT-P-54 b_2020-01-22.png (SHA-256 e1a7b3ea5524f55c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [249, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8f58672ae52d9d02...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), HATP-54200123052113.FITS → HATP-54200123090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-200123.log (14a150d60be6...), FinalLightCurve_HAT-P-54 b_2020-01-22.png (e1a7b3ea5524...), FinalLightCurve_HAT-P-54 b_2020-01-22.csv (5db32b2ab2c4...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:03Z.